

ARES: Extreme Conditions racing

Team members: Suhani Mathur, Shaunabh Bose, Kareem Hassan, Farah Moussa

1. Background

Autonomous driving systems rely heavily on perception and control modules that are typically developed and tested under ideal conditions. However, real-world driving environments introduce significant variability, particularly under adverse weather conditions such as rain. Rain affects multiple components of the system simultaneously, including reduced visibility for cameras, signal degradation for LiDAR and radar, and decreased road traction affecting vehicle stability.

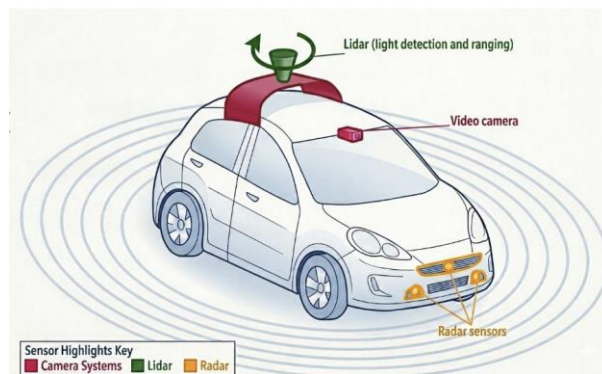


Fig 1: Typical Sensor usage and placement

This project was chosen to address the gap between idealized testing environments and real-world driving conditions. As highlighted in our system design, autonomous vehicles struggle under adverse weather due to degraded sensor inputs, increased noise, and reduced reliability in perception outputs. While previous work often isolates perception or control, this project focuses on integrating both aspects within a unified simulation framework.

Additionally, the project builds upon prior experience in simulation, perception, and system-level engineering, making it possible to explore the interaction between degraded sensor data and vehicle dynamics. By shifting the focus from racing-specific scenarios to high-speed autonomous driving under rain conditions, the project became more aligned with real-world applications and feasible within the scope of the semester.

This project is useful to the world, as it addresses a critical limitation in current autonomous vehicle systems. Adverse weather remains one of the primary barriers to reliable deployment, as it impacts both perception accuracy and vehicle control. Failures in these systems can lead to unsafe driving behavior, especially in high-speed scenarios such as highway environments.



Fig 2: Tesla set in autopilot

The simulation framework developed in this project enables controlled and repeatable evaluation of autonomous driving performance under rain conditions. By modeling both sensor degradation and physical vehicle behavior, the system provides a more realistic testbed compared to traditional simulations that focus only on visual environments.

This approach reduces reliance on expensive and potentially unsafe real-world testing while allowing engineers to systematically analyze how performance degrades under different conditions. Ultimately, this contributes to improving the robustness, safety, and reliability of autonomous driving systems, supporting their broader adoption in real-world transportation.

Autonomous systems have already demonstrated limitations under real-world conditions, particularly when perception systems fail in unexpected environments. For example, in 2018, an autonomous Uber vehicle failed to detect a pedestrian crossing the road at night, leading to a fatal accident. [Investigations](#) showed that the system struggled with perception classification and failed to respond appropriately under low-visibility conditions.

Similarly, Tesla's Autopilot system has been involved in [crashes](#) where the vehicle failed to correctly interpret environmental conditions or detect obstacles, often due to limitations in perception under complex or degraded scenarios.

While these failures are not always directly caused by weather alone, they highlight how degraded perception and environmental uncertainty can lead to critical safety risks, reinforcing the importance of evaluating autonomous systems under adverse conditions such as rain.

These cases highlight the importance of evaluating autonomous systems under non-ideal conditions, where sensor degradation and environmental factors significantly impact system performance. This reinforces the need for controlled simulation environments, such as the one developed in this project, to systematically study failure modes under adverse weather.

Significant research has been conducted in autonomous vehicle perception, simulation, and sensor modeling. Many simulation platforms, such as CARLA and Unity-based environments, have been used to evaluate autonomous systems in controlled settings. However, most of

these environments focus on ideal or near-ideal conditions and do not fully capture the effects of adverse weather.

Recent studies have shown that weather conditions such as rain and fog significantly degrade sensor performance. Camera systems experience reduced visibility, blur, and contrast loss, while LiDAR sensors suffer from point dropout and reduced range. Radar systems, although more robust, introduce increased noise and lower resolution under certain conditions. These effects directly impact tasks such as object detection and lane recognition, reducing overall system reliability.

To address these challenges, multi-sensor approaches have been developed, combining camera, LiDAR, and radar data to improve robustness. However, accurately modeling these sensor interactions within simulation remains a challenge. Most existing work focuses on perception alone without integrating the physical effects of adverse conditions on vehicle dynamics.

Sensor	Strength	Failure in Adverse Weather
Camera	Rich visual detail, object recognition	Reduced visibility, blur, low contrast
LiDAR	Accurate 3D depth, precise geometry	Sparse/noisy points, reflections
Radar	Robust to rain/fog, measures velocity	Low resolution, limited object detail

Fig 3: Sensor comparison table

In parallel, research on vehicle dynamics highlights the importance of modeling tire-road interaction, particularly under wet conditions. Reduced friction leads to increased slip, instability, and changes in vehicle response, especially at high speeds. Despite this, many simulation studies treat perception and vehicle dynamics as separate problems.

This project builds upon these areas by integrating both perception degradation and physics-based vehicle modeling into a single framework. As illustrated in the project workflow, the system combines a computer vision pipeline that generates degraded multi-sensor outputs with a Unity-based simulation environment that models vehicle behavior under wet conditions using Edy's Vehicle Physics.

[Recent work](#) has emphasized the importance of realistic simulation environments for autonomous driving research. Platforms such as CARLA provide high-fidelity simulation with integrated sensor models, enabling testing of perception and control systems under controlled conditions (Dosovitskiy et al., 2017).

However, many existing simulation frameworks still lack detailed modeling of adverse weather effects on both perception and vehicle dynamics. This limitation motivates the development of simulation pipelines that incorporate sensor degradation and physics-based modeling, as explored in this project.

By enabling controlled, parameterized experiments that vary rain intensity and road conditions, this work provides a unified platform for evaluating how autonomous driving performance degrades across perception and control. This integrated approach addresses a key gap in existing research and aligns simulation more closely with real-world driving challenges.

2. Project flow and design

The overall system was designed as a modular pipeline with three main components working together:

1. Computer Vision-based sensor degradation
2. Simulation environment (Unity-based)
3. SLAM and sensor fusion pipeline (ROS2-based)

The flow begins with clean multi-sensor data (camera, LiDAR, radar), which is then artificially degraded using the computer vision pipeline to simulate rain effects. This degraded data is passed into the simulation environment, where vehicle dynamics under wet conditions are modeled. Finally, the SLAM and sensor fusion system processes this data to estimate vehicle localization and evaluate performance under adverse conditions.

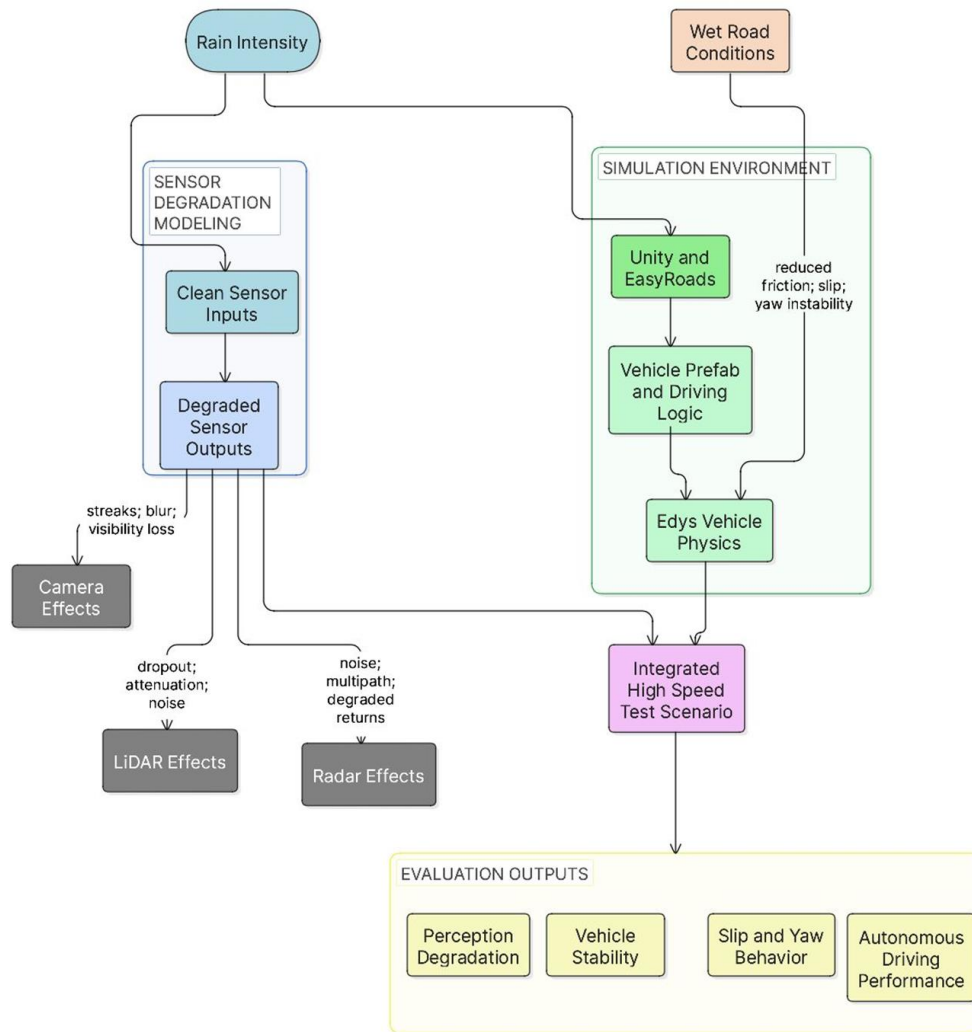


Fig 4: Project Flow Diagram

Initially, the project aimed to build a tightly integrated system combining SLAM, simulation, and data degradation into a single pipeline. However, this approach created strong dependencies between components and slowed development.

To address this, the team redesigned the system into a modular framework, allowing each component to be developed and tested independently. This significantly improved flexibility and made it easier to isolate issues and validate individual subsystems.

The project was then developed step by step:

- **Step 1: Sensor Degradation Modeling**

A computer vision pipeline was developed to simulate rain effects on sensor data. For camera inputs, this included adding rain streaks, blur, reduced contrast, and noise. For LiDAR, point cloud degradation was simulated through random point dropout and

spatial noise. Radar data was incorporated to maintain robustness under noisy conditions.

- **Step 2: Simulation Environment Development**

A physics-based simulation environment was built in Unity. This environment models realistic vehicle behavior under wet conditions, including reduced tire-road friction, lateral slip, and instability. Edy's Vehicle Physics was used to accurately simulate these dynamics.

- **Step 3: SLAM and Sensor Fusion**

A SLAM and odometry pipeline was implemented in ROS2 using multi-sensor fusion. Techniques such as Extended Kalman Filter (EKF) and Factor Graph Optimization (FGO) were used to combine LiDAR, radar, and IMU data for improved localization accuracy, especially under degraded conditions.

- **Step 4: System Evaluation**

Controlled experiments were designed to evaluate system performance. These included comparing clean vs degraded sensor inputs, testing different rain intensities, and analyzing localization accuracy and vehicle stability.

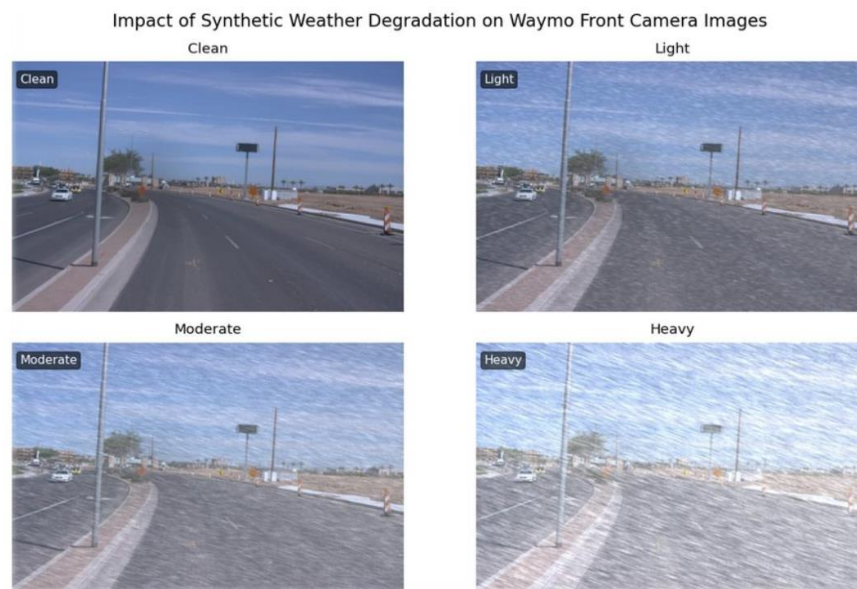


Fig 5: Example of camera degradation with parameters

3. Teammate Contributions

- **Farah Moussa (Computer Vision & Sensor Degradation)**

Led the development of the computer vision pipeline for simulating adverse weather effects on sensor data, focusing on degrading camera inputs through rain streaks, blur, noise, and reduced visibility. Adapted this pipeline to the Waymo Open Dataset by

scaling it from individual images to batch processing thousands of frames, making it more realistic and scalable. Introduced parameterized weather levels (light, moderate, heavy) for controlled experimentation and developed simple evaluation metrics such as contrast and intensity to quantify degradation. Also contributed to modeling the impact of weather on LiDAR and supported integrating multi-sensor degradation into the overall system.

- **Kareem Hassan (Radar Systems & Sensor Analysis)**

Focused on the integration and analysis of radar sensing within the system. Developed a strong foundation in radar signal processing, noise handling, and feature extraction, and explored 4D radar datasets to understand how Doppler velocity and elevation data improve detection under adverse conditions. Investigated methods for extracting useful features such as motion consistency and object density, and experimented with preprocessing techniques including filtering and visualization of radar point clouds. Contributed to system design by evaluating how radar enhances robustness in low-visibility scenarios and supports sensor fusion.

- **Suhani Mathur (Simulation & Vehicle Dynamics)**

Led the development of the Unity-based simulation environment, building a physics-based testbed for evaluating autonomous driving under adverse weather conditions. Implemented realistic vehicle dynamics using Edy's Vehicle Physics, modeling effects such as reduced tire-road friction, lateral slip, and instability on wet surfaces. Focused on creating a controlled and repeatable environment for testing system performance, and ensured that simulation outputs align with real-world driving behavior under rain conditions.

- **Shaunabh Bose (Simulation Support & Integration)**

Assisted in the development and testing of the simulation environment, supporting implementation and validation of system components. Contributed to environment setup, debugging, and refining simulation behavior to ensure consistency and reliability. Worked alongside the simulation lead to help integrate different modules and ensure that sensor inputs and vehicle dynamics operate correctly within the testbed. Also supported testing and evaluation efforts to verify system functionality.

4. Results

At the current stage, the project has successfully established the core components of the framework, with the simulation environment nearing completion and the sensor degradation pipeline implemented.

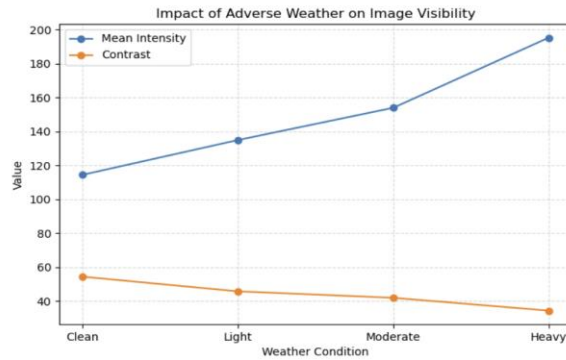


Fig 6: Observations from camera degradation

Key outcomes include

- A working simulation environment capable of modeling realistic vehicle behavior under wet conditions
- A computer vision pipeline for generating degraded sensor data, designed to support multi-sensor inputs under adverse weather conditions
- A SLAM framework designed for multi-sensor fusion under noisy inputs

While full end-to-end integration is still in progress, the system already demonstrates the ability to simulate and analyze how adverse weather impacts sensor performance and vehicle localization.

Preliminary observations show that:

- Camera data is highly affected by visibility loss and noise
- LiDAR suffers from sparsity and measurement errors
- Radar remains more robust but lacks detailed resolution

These findings reinforce the importance of sensor fusion in maintaining reliable autonomous driving performance under adverse conditions.

Overall, the project has not yet reached full completion, but it has successfully built a strong foundation for evaluating autonomous systems in controlled, repeatable adverse weather scenarios.

In terms of the simulation, here are the [photos](#), and different scenarios/ and the progressing steps that build to leading the final environment and deliverable.

5. Future Work

Reflect/Critique & Limitations:

A key part of our approach this semester was redefining the project so it better reflects real-world autonomous driving conditions. On the computer vision side, adapting the rain

corruption pipeline to work on the [Waymo dataset](#) was an important step. Instead of working on simple test images, we were [processing thousands of real driving frames](#), which made the results more meaningful and exposed practical challenges like different lighting conditions and scene variations. Adding simple evaluation metrics like contrast and intensity (see Fig. 6) also helped make the results more structured instead of relying only on visual comparison.

Another thing that improved the project was how the overall direction became clearer over time. Early in the semester, we spent a lot of time deciding what the project should focus on, which led to switching from CARLA to Unity. CARLA required more GPU resources than our laptops could handle, so it was not practical for us to use. Unity turned out to be a better fit, especially because it includes tools like Edy's Vehicle Physics, which made it easier to model physical effects like reduced traction on wet roads. Once we made this decision, it was easier to connect the different parts of the system.

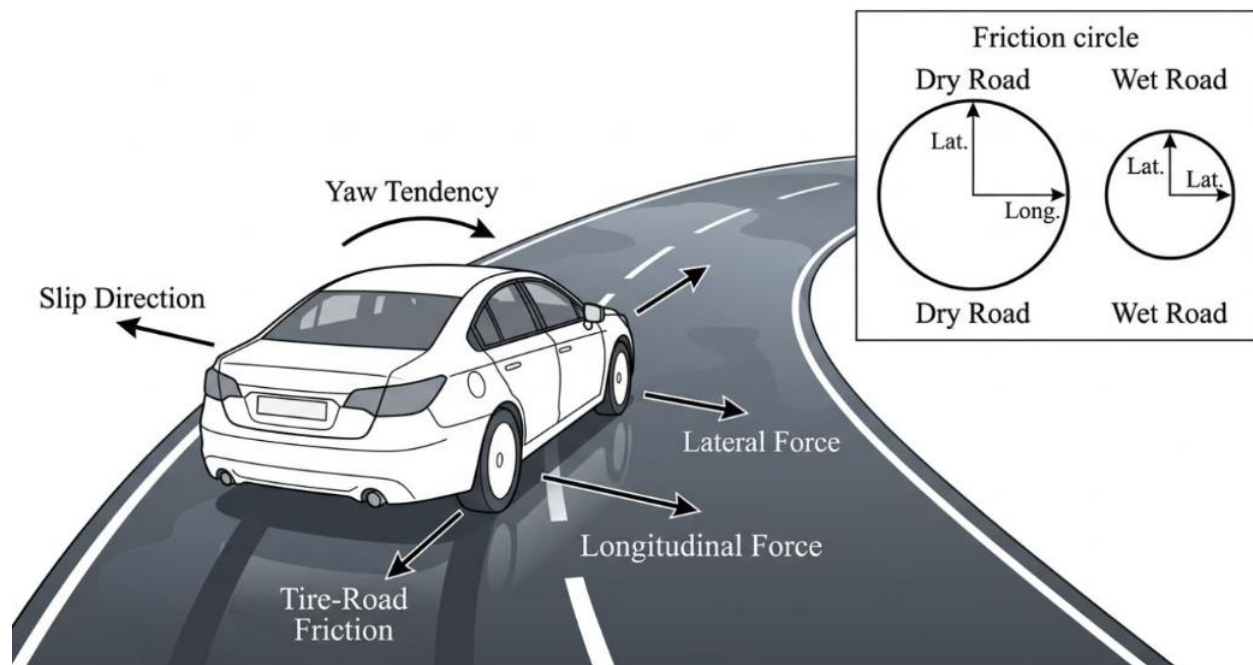


Fig 6: Variables associated with tire slip

We also approached the problem from a multi-sensor perspective by considering camera, LiDAR, and radar together. Each sensor is affected differently by rain, so this helped us think about the system more realistically. For example, camera data is affected by visibility and blur, LiDAR by missing points and noise, while radar remains more reliable in rain but provides less detailed information. This made it clear why combining multiple sensors is important for autonomous systems, especially under degraded conditions.

However, there are still areas that could have been improved. One issue is that it took time early in the semester to clearly define the project direction, which delayed some technical progress and required us to adjust decisions later, such as changing the simulation environment.

Another limitation is related to working with a large dataset like Waymo. We did not fully consider storage and processing requirements at the beginning. Handling thousands of images and generating multiple corrupted versions for different rain intensities quickly became resource-intensive. This made the pipeline harder to manage and slowed down experimentation, especially when trying to test multiple conditions consistently.

There are also challenges in keeping the different components of the system aligned. Since we are working with computer vision, simulation, and multiple sensors at the same time, it is not always straightforward to ensure that everything represents the same conditions. For example, the way rain intensity is applied to images does not always directly match how it should affect LiDAR or the physical simulation, which can make the system less consistent overall.

In terms of the approach itself, the current degradation models are useful for controlled testing, but they simplify some aspects of real-world behavior. For example, the camera pipeline focuses on visual effects like streaks and contrast changes, LiDAR is modeled using dropout and noise, and radar is mainly treated as a more stable reference sensor rather than being fully modeled in detail. These simplifications made it possible to build and test the system within our constraints, but they also limit how closely the results match real-world conditions.

Future Work:

In future semesters, the project can be expanded by improving the scalability of the pipeline so it can handle larger portions of the dataset and support more experiments efficiently. This would allow testing across more scenarios and different levels of rain intensity in a more consistent way.

Another key step is to fully integrate the sensor degradation pipeline with the SLAM and sensor fusion system. Since the simulation already accounts for physical effects of rain such as reduced tire-road friction and vehicle instability, this integration would allow degraded camera, LiDAR, and radar data to be evaluated together with realistic vehicle behavior to study localization and overall system performance under different conditions.

Once these components are fully integrated into the base simulator, future work will expand the framework to model additional extreme weather conditions such as sandstorms and active snowfall. Incorporating these environments would broaden the applicability of the project beyond rain scenarios, enabling testing for Baja, rally, and other off-road racing applications where vehicles must operate reliably under harsh and unpredictable terrain and weather conditions.

By connecting these components, the framework can be used as a controlled testbed where different SLAM or sensor fusion approaches can be compared under the same conditions. This would make it useful not only for our team, but also for other researchers or teams who want to test how their algorithms perform in adverse weather without relying on real-world data collection.